Case Study Rubric – GDP per capita vs. CO_2 emissions

DS 4002 – Instructor: Javier Rasero
Due: May 6, 2026
Submission format: GitHub repository (submitted by link to canvas)
Access the GitHub here:

Why am I doing this? This case study is designed to simulate a real-world data science task at the intersection of economics, environmental science, and policy. Understanding the relationship between GDP per capita and CO_2 emissions is critical for evaluating sustainable development and informing climate policy decisions. By completing this project, you will gain experience working with real-world datasets, identifying meaningful relationships between variables, and building forecasting models that can inform forward-looking decisions.

- Course Learning Objective: Develop and implement a complete data science pipeline, from data analysis to forecasting, and communicate findings effectively.

What am I going to do? You will investigate the relationship between GDP per capita and CO_2 emissions per capita for selected countries using historical data.

Your work will include:
- Cleaning and preparing the dataset for analysis
- Performing exploratory data analysis to identify patterns and trends
- Evaluating whether a relationship exists between GDP per capita and CO_2 emissions per capita, and whether it varies across countries
- Building and applying SARIMA and ARIMAX models to forecast future values of both variables
- Interpreting and communicating your findings through visualizations and written analysis

Your final product should resemble a professional data analysis project that combines statistical insight, modeling, and clear communication.

Tips for success:
- Start early—time-series modeling and debugging can take longer than expected
- Focus on understanding the data before modeling; strong EDA will guide better forecasts
- Be thoughtful in model selection and parameter tuning rather than trying random combinations
- Clearly label and document all visualizations so they are interpretable to a non-technical audience
- Keep your code organized and well-commented to ensure reproducibility
- Use GitHub consistently (commits, structure, readability matter)

How will I know I have succeeded? You will meet expectations on this Case Study when you follow the criteria in the rubric below.

Formatting	● One Github Repository (submitted via link on canvas) ○ A README.md file ○ A LICENSE.md file ○ A SCRIPTS folder ■ Contains all scripts used for analysis ○ A DATA folder ■ Contains original data files, as well as any cleaned or processed versions ○ An OUTPUT folder ■ Contains any images/output files generated during the study
SCRIPTS folder	● Contains well documented python or Jupyter Notebook file(s) with code that executed the EDA and both models. ● Indicate the order in which files should be run, either by including it in the title of the file or the README.md file
OUTPUT folder	● Contains all of the output generated by your project, e.g. figures, tables, etc. ● Importantly, any information like tables, figures shown in your presentation should be here. ● Use informative names for your files.
README.md	● The README should act as a clear guide to your repository so that any user can quickly understand and reproduce your work. It must include three main sections: ● Software and Platform ○ List the programming language/software used, required packages, and the operating system used. ● Repository Structure ○ Provide a clear folder tree showing all directories and files so users understand how the project is organized. ● Reproduction Instructions ○ Give step-by-step, plain-language instructions for how to run the code and reproduce your results, in the correct order.
References	● All references should be listed at the end of the document ● Use IEEE Documentation style (link)